Gitte Kremling

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EDUCATION

Ph.D. in Mathematics Aug 2024
University of Wisconsin-Milwaukee, USA
Thesis: "Bootstrap-based goodness-of-fit test for parametric families of conditional distributions"
Supervised by: Gerhard Dikta and Richard Stockbridge
Graduated with highest possible GPA (4.0)

M.Sc. in Applied Mathematics & Computer Science (formerly Technomathematics) Aug 2019
FH Aachen University of Applied Sciences, Germany
Thesis: "Convergence analysis for multi-level spectral deferred corrections"
Graduated with highest possible GPA (1.0)

B.Sc. in Applied Mathematics & Computer Science (formerly Scientific Programming) Aug 2017
FH Aachen University of Applied Sciences, Germany
Graduated with highest possible GPA (1.0)

WORK EXPERIENCE

Postdoctoral Researcher Apr 2025 – present
University of Hamburg, Germany
Department of Mathematics, PI: Johannes Lederer

Research Associate Sep 2022 – Mar 2025
FH Aachen University of Applied Sciences, Germany
Department of Medical Engineering and Technomathematics

Teaching Assistant Jan 2020 – Jul 2022
University of Wisconsin-Milwaukee, USA
Department of Mathematical Sciences

Research Associate Sep 2017 - Dec 2019
Forschungszentrum Jülich GmbH, Germany
Jülich Supercomputing Center, Division Mathematics and Education

Professional Trainee (Mathematical-Technical Software Developer) Sep 2014 - Aug 2017
Forschungszentrum Jülich GmbH, Germany
Jülich Supercomputing Center, Division Communication Systems

PUBLICATIONS AND PREPRINTS

G. Kremling, J. Näf and J. Lederer (2026, under review). Generative Modeling under Non-Monotonic MAR Missingness via Approximate Wasserstein Gradient Flows. Submitted for publication. arXiv:2604.04567

G. Kremling, F. Iafrate, M. Taheri and J. Lederer (2025, under review). Non-asymptotic error bounds for probability flow ODEs under weak log-concavity. Submitted for publication. arXiv:2510.17608

G. Kremling and G. Dikta (2025, under review). Asymptotic properties of the MLE in distributional regression under random censoring. Submitted for publication. arXiv:2503.14311

G. Kremling and G. Dikta (2025). Bootstrap-based goodness-of-fit test for parametric families of conditional distributions. *Computational Statistics and Data Analysis, 215 (to appear in print March 2026)*, doi:10.1016/j.csda.2025.108289

G. Kremling and R. Speck (2021). Convergence of multilevel spectral deferred corrections. *Communications in Applied Mathematics and Computational Science, 16(2), 227-265*, doi:10.2140/camcos.2021.16.227

SOFTWARE

G. Kremling (2026). FLOWGEM: FLOW-based GEeneration for Missing data [Python code] Available at: https://github.com/gkremling/FLOWGEM.

G. Kremling (2024). gofreg: Bootstrap-Based Goodness-of-Fit Tests for Parametric Regression [R package] Available at: https://CRAN.R-project.org/package=gofreg.

PRESENTATIONS

Statistical Laboratory, University of Cambridge, Cambridge, UK (Invited Seminar Talk) Apr 2026

Workshop on Stochastic Models, Statistics and Their Applications, Würzburg, Germany (Invited Talk) Mar 2026

International Conference on Statistics and Data Science, Seville, Spain (Invited Talk) Dec 2025

Computational Imaging Group, DESY, Hamburg, Germany (Invited Seminar Talk) Nov 2025

Conference on Mathematics of Machine Learning, Hamburg, Germany (Poster) Sep 2025

German Probability and Statistics Days, Dresden, Germany (Talk) Mar 2025

Joint Statistical Meetings, Portland, OR, USA (Talk) Aug 2024

European Meeting of Statisticians, Warsaw, Poland (Talk) Jul 2023

German Probability and Statistics Days, Essen, Germany (Poster) Mar 2023

Graduate Student Colloquium, Department of Mathematical Sciences, University of Wisconsin-Milwaukee, USA (Talks) Dec 2021 & Apr 2022

International Conference on Scientific Computing and Differential Equations, Innsbruck, Austria (Talk) Jul 2019

TEACHING EXPERIENCE

As Instructor

Statistical Modeling (graduate course), FH Aachen	Fall 2024
Algebra (Upward Bound Math & Science program), UW-Milwaukee	Summer 2022
Precalculus (Upward Bound Math & Science program), UW-Milwaukee	Summer 2022
Calculus and Analytic Geometry I (undergraduate course), UW-Milwaukee	Spring 2022
Algebraic Literacy I and II (undergraduate course), UW-Milwaukee	Fall 2021
Precalculus (undergraduate course), UW-Milwaukee	Spring 2021
Introduction to College Algebra (undergraduate course), UW-Milwaukee	Fall 2020

As Teaching Assistant

Fundamental Concepts of Arithmetic (undergraduate course), Uni Hamburg	Spring 2026
Introduction to Mathematical Thinking and Methods (undergraduate course), Uni Hamburg	Fall 2025
Functional Analysis (graduate course), FH Aachen	Fall 2024
Numerical Methods in Machine Learning (graduate course), FH Aachen	Fall 2024
Bootstrap Methods in Statistics (graduate course), FH Aachen	Spring 2024
Statistical Modeling (graduate course), FH Aachen	Fall 2023
Time Series Analysis and Prognosis (graduate course), FH Aachen	Fall 2023
Numerical Analysis (undergraduate course), FH Aachen	Spring 2019 & 2023
Probability Theory (undergraduate course), FH Aachen	Fall 2019

SERVICE

Member of Faculty Council, Faculty of Mathematics, Informatics and Natural Sciences, University of Hamburg — Oct 2025 – present

Postdoc Network Spokesperson, Hamburg Research Academy — Oct 2025 – present

WORKSHOPS AND SUMMER SCHOOLS

Workshop on Mathematical Data Science, Black Forest, Germany — Jul 2025
Led an open-question session on white-box transformers and contributed to individual and group discussions throughout the workshop

Mathematical Problems in Industry Workshop, Worcester, MA, USA — Jun 2022
Collaborated with a team to develop innovative approaches for regression analysis of sparse and noisy data

Graduate Student Mathematical Modeling Camp, Newark, DE, USA — Jun 2022
Contributed to the development of a model for the dynamics of a snowflake in a cloud microphysics framework

HONORS AND AWARDS

Badge of Honor (M.Sc.) — Aug 2019
FH Aachen University of Applied Sciences, Germany
Awarded to the best 5% of graduates

Deutschlandstipendium — Sep 2018 - Aug 2019
FH Aachen University of Applied Sciences, Germany
Provides financial and non-material support to high-achieving and committed students

Badge of Honor (B.Sc.) — Aug 2017
FH Aachen University of Applied Sciences, Germany
Awarded to the best 5% of graduates

PROFESSIONAL MEMBERSHIPS

American Statistical Association
Caucus for Women in Statistics and Data Science
DMV Fachgruppe Stochastik
Institute of Mathematical Statistics

REFERENCES

Johannes Lederer, Postdoctoral Advisor
University of Hamburg, Germany
Homepage johannes.lederer@uni-hamburg.de

Gerhard Dikta, PhD Co-Advisor
FH Aachen University of Applied Sciences, Germany
Homepage dikta@fh-aachen.de

Richard Stockbridge, PhD Co-Advisor
University of Wisconsin-Milwaukee, USA
Homepage stockbri@uwm.edu

Hayley Nathan, Course Coordinator
University of Wisconsin-Milwaukee, USA
Homepage henathan@uwm.edu

Robert Speck, Master's Thesis Advisor
Forschungszentrum Jülich, Germany
Homepage r.speck@fz-juelich.de